From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

PART TEST – IV

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

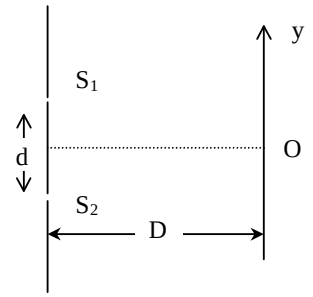
Single Correct Choice Type

This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Two mirrors, one concave and the other convex, are placed 60 cm apart with their reflecting surfaces facing each other. An object is placed 30 cm from the pole of either of them on their axis. If the focal lengths of both the mirrors are 15cm, the position of the image formed by reflection, first at the convex and then at the concave mirror, is
(A) 19.09 cm from the pole of the convex mirror
(B) 19.09 cm from the pole of the concave mirror
(C) 11.09 cm from the pole of the concave mirror
(D) 11.09 cm from the pole of the convex mirror

2. An alternating voltage $E = E_0 \sin \omega t$, is applied across a coil of inductor L. The current flowing through the circuit at any instant t is
(A) $\frac{E_0}{\omega L} \sin(\omega t + \pi/2)$ (B) $\frac{E_0}{\omega L} \sin(\omega t - \pi/2)$
(C) $E_0 \omega L \sin(\omega t - \pi/2)$ (D) $E_0 \omega L \sin(\omega t + \pi/2)$

3. Consider a usual set up of Young's double slit experiment with slits of equal intensity as shown in the figure. Take 'O' as origin and the Y axis as indicated. If average intensity between $y_1 = -\frac{\lambda D}{4d}$ and $y_2 = +\frac{\lambda D}{4d}$ equals 'n' times the intensity of maxima, then 'n' equals



(A) $\frac{1}{2}\left(1+\frac{2}{\pi}\right)$ (B) $2\left(1+\frac{2}{\pi}\right)$
(C) $\left(1+\frac{2}{\pi}\right)$ (D) $\frac{1}{2}\left(1-\frac{2}{\pi}\right)$

Space for rough work

4. One of the combination from the fundamental physical constants is hc/G. The unit of this expression is
(A) kg^2 (B) m^3
(C) s^{-1} (D) none of the above

5. Let $[\varepsilon_0]$ denote the dimensional formula of the permittivity of vacuum. If M = mass, L = length, T = time and I = electric current. Then
(A) $[\varepsilon_0] = [M^{-1}L^{-3}T^2I]$ (B) $[\varepsilon_0] = [M^{-1}L^{-3}T^4I^2]$
(C) $[\varepsilon_0] = [M^{-1}L^2T^{-1}I^{-2}]$ (D) $[\varepsilon_0] = [M^{-1}L^2T^{-1}I]$

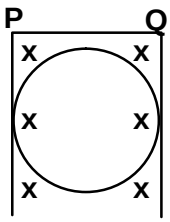
6. A coil of resistance R and inductance L is connected across an a.c. power supply of r.m.s. voltage V. The average power dissipated in the coil is,
(A) V^2/R (B) $\dfrac{V^2R}{(R^2+\omega^2L^2)}$
(C) $\dfrac{V^2}{\sqrt{(R^2+\omega^2L^2)}}$ (D) zero

7. The figure shows a plane wave front at a time 't' and at a time 't_1'
In the time interval (t_1 – t) the wave front must have passed through



(A) a prism (B) a prism and a convex lens
(C) a convex lens (D) a plane mirror and a concave lens

8. A vertical ring of radius r and resistance R falls vertically. It is in contact with two vertical rails which are joined at the top. The rails are without friction and resistance. There is a horizontal uniform magnetic field of magnitude B perpendicular to the plane of the ring and the rails. When the speed of the ring is v, the current in the section PQ is

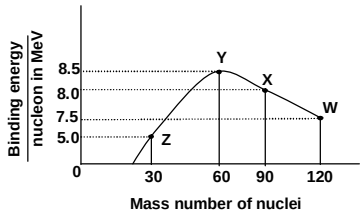


(A) zero (B) $\dfrac{2Brv}{R}$
(C) $\dfrac{4Brv}{R}$ (D) $\dfrac{8Brv}{R}$

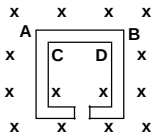
Space for rough work

9. An electric lamp designed for operation on 110 V AC is connected to a 220 V AC supply, through a choke coil of inductance 2H, for proper operation. The angular frequency of the AC is $100\sqrt{10}$ rad/s. If a capacitor is to be used in place of the choke coil, its capacitance must be
(A) 1 μF (B) 2 μF
(C) 5 μF (D) 10 μF

10. Binding energy per nucleon v_s mass number curve for nuclei is shown in the figure W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is
(A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$
(C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$



11. A wire is bent to form the double loop shown in the figure. There is a uniform magnetic field directed into the plane of the loop. If the magnitude of this field is decreasing, current will flow from
(A) A to B and C to D (B) B to A and D to C
(C) A to B and D to C (D) B to A and C to D



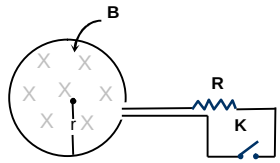
12. A cubical room is formed with 6 plane mirrors. An insect moves along the diagonal of the floor with uniform speed. The speed of its image in two adjacent walls is $20\sqrt{2}$ cm/s. Then the speed of the image formed on the roof is
(A) 20 cm/s (B) 40 cm/s
(C) $20\sqrt{2}$ cm/s (D) $10\sqrt{2}$ cm/s

13. A prism of refractive index $\sqrt{2}$ refracting angle A produces minimum deviation D_m of a ray on one face at an angle of incidence 45° . The values of A and D_m are respectively.
(A) 45°, 45° (B) 45°, 60°
(C) 60°, 30° (D) 60°, 45°

Space for rough work

14. Which of the following statement is correct?
(A) The rest mass of a stable nucleus is less than the sum of the rest masses of its separated nucleons.
(B) The rest mass of a stable nucleus is greater than the sum of the rest masses of separated nucleons.
(C) In nuclear fission, energy is released by fusing two nuclei of medium mass (approximately 100 amu)
(D) In nuclear fission, energy is released by fragmentation of a very heavy nucleus.

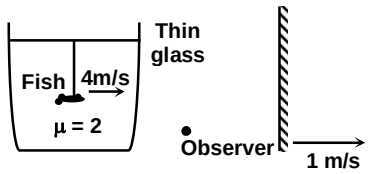
15. Shown in the figure is a circular loop of radius r and resistance R. A variable magnetic field of induction $B = e^{-t}$ is established inside the coil. If the key (K) is closed at t = 0, the electrical power developed at the instant is equal to



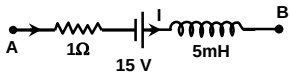
(A) $\frac{\pi r^2}{R}$ (B) $\frac{10r^3}{R}$
(C) $\frac{\pi^2 r^4 R}{5}$ (D) $\frac{\pi^2 r^4}{4R}$

16. In a YDSE apparatus, two films of thickness t_A and t_B, having refractive indices μ_A and μ_B are placed in front of slits A and B respectively. If $t_A\mu_A = t_B\mu_B$ the central maximum will
(A) not shift (B) shift towards A if $t_A < t_B$
(C) shift towards B if $t_A < t_B$ (D) shift towards A if $t_A > t_B$

17. Velocity of image of fish in the mirror with respect to ground is
(A) zero
(B) 2 m/s towards mirror.
(C) 2 m/s away from mirror
(D) 4 m/s towards mirror.



18. The network shown in figure is a part of a complete circuit. If at a certain instant the current (I) is 5A, and is decreasing at a rate of 10^3 A/s. Then $V_B - V_A$ is equal to



(A) 15V (B) 25V
(C) 20V (D) 10 V

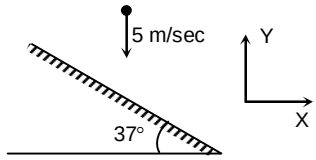
Space for rough work

19. Nuclei of radioactive element A are produced at any time t. the element A has decay constant λ. Let N be number of nuclei if element A at any time t. At time $t = t_0$, dN/dt is minimum. Then the numbers of nuclei of element A at time $t = t_0$ is

(A) $\frac{t_0 - \lambda t_0^2}{\lambda^2}$ (B) $\frac{2t_0 - \lambda t_0^2}{\lambda^2}$

(C) $\frac{t_0 - \lambda t_0^2}{\lambda}$ (D) $\frac{2t_0 - \lambda t_0^2}{\lambda}$

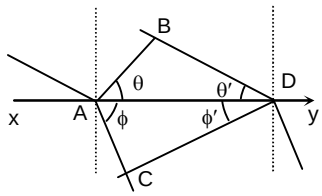
20. An object O is just about to strike a perfectly reflecting inclined plane of inclination 37° its velocity is 5 m/sec. Find the velocity of its image.



(A) $3\hat{i} + 4\hat{j}$ (B) $4\hat{i} + 3\hat{j}$

(C) $4.8\hat{i} + 14\hat{j}$ (D) $1.4\hat{i} + 4.8\hat{j}$

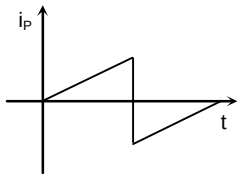
21. A wave front AB moving in air is incident on a plane glass surface XY, as given in the figure, its position CD after refraction through glass slab is shown along with normal drawn at A and d. the refractive index of glass w.r.t. air is given by



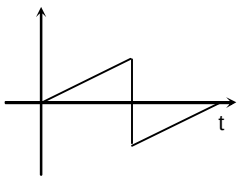
(A) $\frac{\sin\theta}{\sin\theta'}$ (B) $\frac{\sin\theta}{\sin\phi'}$

(C) $\frac{\sin\phi}{\sin\theta}$ (D) $\frac{AB}{CD}$

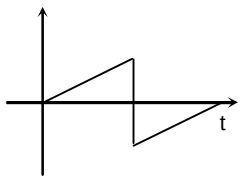
22. Nature of induced emf in secondary coil due to current in primary coil.



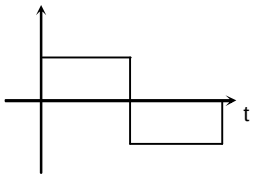
(A)



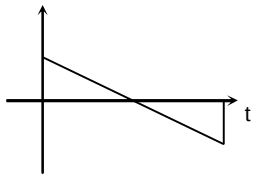
(B)



(C)

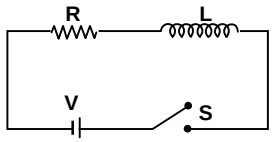


(D)



Space for rough work

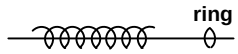
23. A resistor and inductor are connected in series through a battery. The switch is closed at time t = 0. Then what is the magnitude of current flowing when rate of increase of magnetic energy in the inductor is maximum



(A) $I = \frac{V}{R}$ (B) $I = \frac{V}{2R}$

(C) $I = \frac{V}{4R}$ (D) $I = \frac{V}{R}\ln 2$

24. The diagram shows a solenoid carrying time varying current $I = I_0 t$ (I_0 is constant) on the axis of this solenoid a conducting ring is being placed as shown in the figure. The mutual inductance of the ring and the solenoid is M and self inductance of the ring is L. If the resistance of the ring is R then the maximum current which can flow through the ring is:



(A) $(2M+L)\frac{I_0}{R}$ (B) $\frac{MI_0}{R}$

(C) $(2M-L)\frac{I_0}{R}$ (D) $(M+L)\frac{I_0}{R}$

25. A student constructs a series RLC circuit. White operating the circuit at a frequency f he uses an AC voltmeter and measures the potential difference across each device as $(\Delta V_R)_{max}$ = 8.8 V, $(\Delta V_L)_{max}$ = 2.6 V, and $(\Delta V_C)_{max}$ = 7.4 V. Inductor in the circuit is connected next to capacitor. What result student will obtain for a measurement of combined potential difference $(\Delta V_L + \Delta V_C)_{max}$ across the inductor and capacitor.

(A) 10.0 V (B) 7.8 V

(C) 7.4V (D) 4.8 V

26. A radioactive isotopes is being produced at a constant rate X. Half life of substance is Y. After same time the no of radioactive nuclei become constant. The value of constant is

(A) $\frac{XY}{\ln(2)}$ (B) XY

(C) (ln2)XY (D) $\frac{X}{Y}$

27. In a black body radiation certain temperature T_1, the wave length having maximum intensity of radiation equal 9000 A^0. when the temperature is increased from T_1 to T_2 the total radiation increases 16 times. The peak radiation at T_2 is found to be capable of ejecting photoelectrons. The maximum kinetic energy of photoelectrons is the same as the energy of photon that one gets when one of electron in the M-shall of hydrogen atom jumps to L-shell, the work function of the metal will be

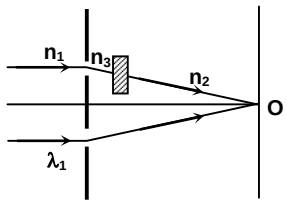
(A) 0.88 eV (B) 1.5 eV

(C) 2.0 eV (D) 1.2 eV

Space for rough work

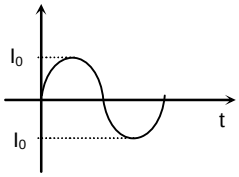
28. The energy of tungsten atom with a vacancy in L shell is 11.3 keV. Wavelength of k_α photon for tungsten is 21.3 pm. If a potential difference of 62 kV is applied across the X-ray tube following characteristic.
(A) K, L series
(B) only k_α and L series
(C) only L series
(D) none of these

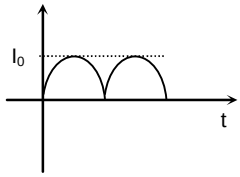
29. In the figure shown YDSE, a parallel beam of light is incident on the slit from a medium of reflective index n_1. The wavelength of light in the medium λ_1. A transparent slab of thickness 't' and refractive index n_3 is put infront of one slit. The medium between the screen and the plane of the slits is n_2. The phase difference between the light waves reaching point O (Symmetrical relative to the slits) is

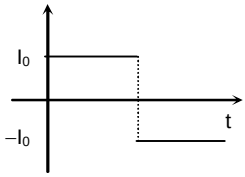


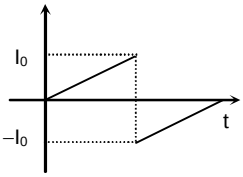
(A) $\frac{2\pi}{n_1\lambda_1}(n_3 - n_2)t$
(B) $\frac{2\pi}{\lambda_1}(n_3 - n_2)t$
(C) $\frac{2\pi n_1}{n_2\lambda_1}\left(\frac{n_3}{n_2} - 1\right)t$
(D) $\frac{2\pi n_1}{\lambda_1}(n_3 - n_1)t$

30. If I_1, I_2, I_3 and I_4 are the respective rms values of the time varying current as shown in four cases I, II, III, IV. Then identify the correct relations:

(I)


(II)


(III)


(IV)


(A) $I_1 = I_2 = I_3 = I_4$
(B) $I_3 > I_1 = I_2 > I_4$
(C) $I_3 > I_4 > I_2 = I_1$
(D) $I_3 > I_2 > I_1 > I_4$

Rough work

Chemistry PART – II

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Before adding the reagents of group (III) of basic radicals, the solution is heated with some concentrated HNO_3 in order to
   (A) lower the pH
   (B) oxidize Cr^{3+} to $Cr_2O_7^{2-}$
   (C) oxidize Fe^{2+} to Fe^{3+}
   (D) increase $[NO_3^-]$

2. When heated above 916°C, iron changes its crystal structure from bcc to ccp structure without any change in the radius of atom. The ratio of density of the crystal before heating and after heating is:
   (A) 1.089
   (B) 0.918
   (C) 0.725
   (D) 1.231

3. The aqueous solution of each of the following salt is coloured except
   (A) $TiCl_4$
   (B) $FeCl_3$
   (C) $CuCl_2$
   (D) $MnCl_2$

4. If all Fe^{2+} ions are replaced by Sr^{4+} ions in 0.1 mole of $Fe_{0.9}O$, then number of vacancies created are
   (A) 0.035 N_A
   (B) 0.01 N_A
   (C) 0.07 N_A
   (D) 0.1 N_A
   (N_A is Avogadro number.)

5. In diamond, the lattice is fcc with C atoms occupying lattice points as well as some of the voids. Which of the following statement(s) is/are correct about the diamond structure?
   (I) In the diamond lattice, half of the tetrahedral voids are occupied.
   (II) Percentage space occupied (packing efficiency) is 74%.
   (III) In the diamond lattice all octahedral voids are occupied.
   (A) Only I
   (B) Only III
   (C) I, II and III
   (D) Only II

Rough work

6. Na_2CO_3 cannot be used in place of $(NH_4)_2CO_3$ for the precipitation of group V cations, because
(A) Na^+ interferes in the detection of group V (B) concentration of CO_3^{2-} is very low
(C) Na^+ will react with the acid radicals (D) Mg^{2+} will precipitate as $MgCO_3$

7. When $K_4[Fe(CN)_6]$ is added to $FeCl_3$, the complex formed is
(A) $Fe_3[Fe(CN)_6]_4$ (B) $K_2Fe[Fe(CN)_6]$
(C) $Fe_4[Fe(CN)_6]_3$ (D) $K_2Fe[Fe(CN)_6]_2$

8. When excess of $SnCl_2$ is added to a solution of $HgCl_2$, a white precipitate that turn grey, is obtained. The grey colour is due to the formation of
(A) Hg_2Cl_2 (B) $SnCl_4$
(C) Sn (D) Hg

9. For an ideal binary liquid solution with $P_A^0 > P_B^0$, which relation between χ_A (mole fraction of A in liquid phase) and Y_A (mole fraction of A in vapour phase) is correct? (χ_B & Y_B are the mole fractions of B in liquid and vapour phase respectively)
(A) $\frac{Y_A}{Y_B} < \frac{\chi_A}{\chi_B}$ (B) $\chi_A > \chi_B$
(C) $Y_A < Y_B$ (D) $\frac{Y_A}{Y_B} > \frac{\chi_A}{\chi_B}$

10. The ratio of van't Hoff factor, if PBr_5 and PCl_5 are assumed to be soluble in water and remains same as in solid state, is
(A) 2 (B) 1
(C) 1.5 (D) 2.5

11. Insulin $(C_2H_{10}O_5)_n$ is dissolved in a suitable solvent and the osmotic pressure (π) of solution of various concentration (Kg/m^3) is measured at 20°C. The slope of a plot of π against C is found to be 4.65×10^{-3} (SI units). The molecular weight of the insulin (in Kg) is
(A) 4.8×10^5 (B) 9×10^5
(C) 3×10^5 (D) 5.24×10^5

12. When 0.1 mole of AgCl is dissolved in 1L of water then relative lowering of vapour pressure is 3.6×10^{-7}. The solubility product of AgCl is
(A) 10^{-10} (B) 2×10^{-10}
(C) 2×10^{-9} (D) Can not calculate

Rough work

13. The weight of a non-volatile solute (molecular weight = 40), required to be dissolved in 114 g octane, to reduce its vapour pressure to 80% is
(A) 5 g (B) 10 g
(C) 15 g (D) 20 g

14. Which is not a π-bonded complex?
(A) Zeises salt (B) Ferrocene
(C) Dibenzene chromium (D) Tetraethyl lead

15. The equilibrium constant of the following redox reaction at 298 K is 1×10^8.
$2Fe^{3+}(aq.) + 2I^-(aq.) \rightleftharpoons 2Fe^{2+}(aq.) + I_2(s)$
If the standard reduction potential of iodine becoming iodide is + 0.54 V. What is the standard reduction potential of Fe^{3+}/Fe^{2+}?
(A) + 1.006 V (B) –1.006 V
(C) + 0.776 V (D) –0.776 V

16. A cell assembled of two hydrogen electrodes at 1 atm pressure. The positive electrode is in contact with a solution of 10^{-4} M hydrogen ions. The emf of the cell is 0.118 V at 25°C. The concentration of hydrogen ions at negative electrode is
(A) 10^{-6} M (B) 10^{-4} M
(C) 10^{-2} M (D) None of the above

17. On passing CrO_2Cl_2 vapours in water and then adding $(CH_3COO)_2Pb$, the precipitate formed is
(A) $PbCrO_4$ (B) $PbCl_2$
(C) $CrCl_3$ (D) $PbSO_4$

18. From the following reaction,
$A^+ + e^- \longrightarrow A \quad E^o = 2$ V
$B^+ + e^- \longrightarrow B \quad E^o = 4$ V
$C^+ + e^- \longrightarrow C \quad E^o = 5$ V
Which cell will have maximum standard electrode potential?
(A) $B|B^+||A^+|A$ (B) $A|A^+||C^+|C$
(C) $A|A^+||B^+|B$ (D) $B|B^+||C^+|C$

19. A constant current of 2 ampere was passed for 16 minutes when 250 ml, 0.2 M $CuSO_4$(aq) solution was electrolysed using a platinum anode and Cu cathode. Then at what temperature the solution will freeze, after passage of current? ($K_f(H_2O)$ = 1.86 unit)
(A) 0.67°C (B) –0.67°C
(C) –1.86°C (D) +1.86°C

Rough work

20. When phosphorus is heated with conc. HNO_3, it reduces the acid to
(A) NO (B) NO_2
(C) N_2O_3 (D) N_2O_5

21. Heating ammonium dichromate yields a gas, which is also obtained by
(A) Mg_3N_2 in H_2O (B) Heating NH_4NO_3
(C) Heating NH_4NO_2 (D) Na_2O_2 in H_2O

22. Which of the following statement is incorrect?
(A) Silver glance mainly contains silver sulphide
(B) Zinc blende mainly contains zinc chloride
(C) Gold is found in native state
(D) Copper pyrites also contains Fe_2S_3

23. In which one case of the following oxides, carbon is not used as a reducing agent (in metallurgical process)?
(A) Fe_2O_3 (B) PbO
(C) SnO_2 (D) Al_2O_3

24. $NH_3 + O_2 \xrightarrow[\Delta]{Pt} X + H_2O$;

$X + O_2 \longrightarrow Y$;

$Y + O_2 + H_2O \longrightarrow Z$;

X, Y and Z are respectively
(A) N_2O, NO_2, HNO_3 (B) NO_2, NO, HNO_3
(C) NO, NO_2, HNO_3 (D) N_2O, NO, HNO_3

25. In the manufacture of steel from pig iron by Bassemer process, the furnace called Baseemer converter is lined with silica bricks when:
(A) the impurity is mainly phosphorus (B) the impurity is mainly manganese
(C) the impurity is mainly silicon (D) the impurity is mainly sulphur

26. To a metal nitrate, when KI solution is added, a brownish black precipitate is produced at first, on adding excess of KI orange solution is produced. Identify the metal ion.
(A) Hg^{2+} (B) Bi^{3+}
(C) Cu^{2+} (D) Pb^{2+}

27. In nitroprusside ion, the iron and NO exist as Fe(II) and NO^+ rather than Fe(III) and NO. These forms can be differentiated by
(A) Estimating the concentration of iron (B) Measuring the concentration of CN^-
(C) Measuring the solid state magnetic moment (D) Thermally decomposing the compound

Rough work

28. The stability of lyophilic colloids is due to
(A) charge on their particles
(B) large size of their particles
(C) smaller size of their particles
(D) a layer of medium of dispersion on their particles

29. Which of the following give HIO_3 on hydrolysis?
(A) ICl_3 (B) IF_5
(C) IF_7 (D) ICl

30. Which of the following is correct about P_4O_{10}?
(A) It is an anhydride of H_3PO_3.
(B) All the P atoms are sp^3d hybridized.
(C) There are 10 (P — O — P) linkage in one molecule.
(D) Total number of sp^3 hybridized atoms per molecules is 10.

Rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. A nine digit number is formed using 1, 2, 3, ….., 9 without repetition. The probability of forming a number such that product of any of its five consecutive digits is divisible by 3 or 5

(A) $\frac{1}{63}$ (B) $\frac{17}{18}$

(C) $\frac{62}{63}$ (D) none of these

2. The area of the region bounded between the curves $y = e^{\|x|\ln|x\|}$, $x^2 + y^2 - 2(|x| + |y|) + 1 \geq 0$ and x-axis where $|x| \leq 1$, if α is the x-coordinate of the point of intersection of curves in 1st quadrant, is

(A) $4\left[\int_0^{\alpha} ex\ln x\,dx + \int_{\alpha}^{1}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$ (B) $4\left[\int_0^{\alpha} ex\ln x\,dx - \int_{1}^{\alpha}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$

(C) $2\left[-\int_0^{\alpha} ex\ln x + \int_{\alpha}^{1}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$ (D) $2\left[\int_0^{\alpha} ex\ln x + \int_{\alpha}^{1}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$

3. If $\bar{a},\bar{b},\bar{c}$ are non-coplanar and $\bar{d}$ is any unit vector the expression $\left|(\bar{a}\cdot\bar{d})(\bar{b}\times\bar{c})+(\bar{b}\cdot\bar{d})(\bar{c}\times\bar{a})+(\bar{c}\cdot\bar{d})(\bar{a}\times\bar{b})\right|$ is independent of the vector

(A) $\bar{a}$ (B) $\bar{b}$

(C) $\bar{c}$ (D) $\bar{d}$

4. If A and B are square matrices such that A^{2006} = O and AB = A + B, then det B equals to

(A) 0 (B) 1

(C) – 1 (D) none of these

Rough work

5. Solution of the differential equation $y(2x^4 + y)\frac{dy}{dx} = (1 - 4xy^2)x^2$ is given by

(A) $3(x^2y)^2 + y^3 - x^3 = c$ (B) $xy^2 + \frac{y^3}{3} - \frac{x^3}{3} + c = 0$

(C) $\frac{2}{5}yx^5 + \frac{y^3}{3} = \frac{x^3}{3} - \frac{4xy^3}{3} + c$ (D) none of these

6. Let A, B and C are any three points on the ellipse $36x^2 + \frac{y^2}{192} = 1$, then the maximum area of the triangle ABC is

(A) 1 (B) 2
(C) 3 (D) 4

7. The length of the perpendicular from (1, 0, 2) on the line $\frac{x+1}{3} = \frac{y-2}{-2} = \frac{z+1}{-1}$ is

(A) $\frac{3\sqrt{6}}{2}$ (B) $\frac{6\sqrt{3}}{5}$

(C) $3\sqrt{2}$ (D) $2\sqrt{3}$

8. The value of the determinant $\begin{vmatrix} 1 & e^{i\pi/3} & e^{i\pi/4} \\ e^{-i\pi/3} & 1 & e^{2i\pi/3} \\ e^{-i\pi/4} & e^{-2i\pi/3} & 1 \end{vmatrix}$ where $i = \sqrt{-1}$ is

(A) $2 + \sqrt{2}$ (B) $-(2+\sqrt{2})$

(C) $-2 + \sqrt{3}$ (D) $2\sqrt{3}$

9. The area of the region containing the points satisfying $|y| + \frac{1}{2} \le e^{-|x|}$, $\max(|x|, |y|) \le 2$ is

(A) $2\log\left(\frac{e}{2}\right)$ (B) $2\log\left(\frac{2e}{3}\right)$

(C) $3\log\left(\frac{e}{2}\right)$ (D) $3\log\left(\frac{2e}{3}\right)$

10. A point P(x, y) moves is such a way that [x + y + 1] = [x] (where [ ] denotes g.i.f) and $x \in (0, 2)$. Then the area representing all the possible positions of P equals

(A) $\sqrt{2}$ sq units (B) $2\sqrt{2}$ sq units

(C) $4\sqrt{2}$ sq units (D) None of these

Rough work

11. If the general solution for the differential equation $\frac{dy}{dx} = \frac{y}{x} + \phi\left(\frac{x}{y}\right)$ is $y = \frac{x}{\ln(cx)}$, then function $\phi\left(\frac{x}{y}\right)$ is

(A) $-\frac{x^2}{y^2}$ (B) $-\frac{y^2}{x^2}$

(C) $\frac{y^2}{x^2}$ (D) $\frac{x^2}{y^2}$

12. The length of the projection of the line segment joining P(–1, 2, 0) and Q(1, – 1, 2) on the plane $2x – y – 2z = 4$ is

(A) 4 (B) 1

(C) 17 (D) none of these

13. Two persons Ram and Shyam agree to meet at a place between 11 to 12 noon. The first one to arrive waits for 20 minute and then leave. If the time of their arrive be independent and at random, then the probability that ram and Shyam meet is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$

(C) $\frac{4}{9}$ (D) $\frac{5}{9}$

14. If one of the roots of the equation $\begin{vmatrix} 7 & 6 & x^2-13 \\ 2 & x^2-13 & 2 \\ x^2-13 & 3 & 7 \end{vmatrix} = 0$ is x = 2, then sum of all other five roots is

(A) $2\sqrt{15}$ (B) – 2

(C) $\sqrt{20}+\sqrt{15}-2$ (D) none of these

15. $\vec{a}$ and $\vec{c}$ are unit vectors and $|\vec{b}| = 4$. If the angle between $\vec{b}$ and $\vec{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$ and $\vec{a}\times\vec{b} = 2\vec{a}\times\vec{c}$ then $\vec{b} = \lambda\vec{a} + 2\vec{c}$ where λ is equal to

(A) $\pm\frac{1}{4}$ (B) $\pm\frac{1}{2}$

(C) ± 1 (D) ± 4

Rough work

16. If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$ and $A^n = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, then minimum value of n is

(A) 2 (B) 4
(C) 6 (D) 3

17. Solution of differential equation $xdy - (y + xy^3(1 + \log x))dx = 0$

(A) $\frac{-x^2}{y^2} = \frac{2x}{3}\left(\frac{2}{3} + \log x\right) + c$ (B) $\frac{x^2}{y^2} = \frac{2x^2}{3}\left(\frac{2}{3} + \log x\right) + c$

(C) $\frac{-x^2}{y^2} = \frac{2x^3}{3}\left(\frac{2}{3} + \log x\right) + c$ (D) none of these

18. If y_1 and y_2 be solution of differential equation $\frac{dy}{dx} + Py = Q$, where P and Q are function of x alone and $y_2 = y_1 z$ then (where a is arbitrary constant)

(A) $z = 1 + ae^{-\int \frac{Q}{y_1} dx}$ (B) $z = 1 - ae^{-\int \frac{Q}{y_1} dx}$

(C) $z = 1 + ae^{\int \frac{Q}{y_1} dx}$ (D) $z = 1 - ae^{\int \frac{Q}{y_1} dx}$

19. If $\vec{p}$, $\vec{q}$ are two non collinear vectors such that $(b-c)\vec{p} \times \vec{q} + (c-a)\vec{p} + (a-b)\vec{q} = 0$, where a, b, c are length of sides of triangle then triangle is

(A) right angled (B) obtuse angled
(C) equilateral (D) isosceles

20. A draws two card at random from a pack of 52 cards. After returning them to the pack and shuffling it, B draws two card at random. The probability that there is exactly one common card is

(A) $\frac{25}{546}$ (B) $\frac{50}{663}$

(C) $\frac{23}{663}$ (D) none of these

21. Area bounded by $y = f^{-1}(x)$ and tangent and normal drawn to it at the points with abscissae π and 2π, where $f(x) = \sin x - x$ is

(A) $\frac{\pi^2}{2} - 1$ (B) $\frac{\pi^2}{2} - 2$

(C) $\frac{\pi^2}{2} - 4$ (D) $\frac{\pi^2}{2}$

Rough work

22. Area enclosed by ellipse $x^2 + \sin^4\alpha\, y^2 = \sin^2\alpha$, $\alpha \in \left(0, \frac{\pi}{2}\right)$ is

(A) 2π (B) π
(C) 1 (D) none of these

23. $\bar{a}, \bar{b}, \bar{c}$ are the edges of a cube of unit length and $\bar{r}$ be any unit vector inside the cube then $|\bar{a} \times \bar{r}|^2 + |\bar{b} \times \bar{r}|^2 + |\bar{c} \times \bar{r}|^2$ is

(A) 0 (B) 1
(C) 2 (D) 3

24. Let X = {1, 2, ,3, ….., 9} and Y = {1, 2, 3…,9} be two sets and one number is selected from each set, then the probability that sum of the numbers selected is 11

(A) $\frac{1}{81}$ (B) $\frac{3}{8}$
(C) $\frac{2}{81}$ (D) $\frac{1}{9}$

25. If $\begin{bmatrix} \frac{1}{25} & 0 \\ x & \frac{1}{25} \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ -a & 5 \end{bmatrix}^{-2}$, then the value of x is

(A) $\frac{a}{125}$ (B) $\frac{2a}{25}$
(C) $\frac{2a}{125}$ (D) none of these

26. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $A^2 + 5A =$

(A) A (B) 2A
(C) 5A (D) 6A

Rough work

27. The solution of differential equation $2x^3y\,dy + (1 - y^2)(x^2y^2 + y^2 - 1)dx = 0$
(A) $x^2y^2 = (cx + 1)(1 - y^2)$ (B) $x^2y^2 = (cx + 1)(1 + y^2)$
(C) $x^2y^2 = (cx - 1)(1 - y^2)$ (D) none of these

28. The numbers of unit vectors perpendicular to $\bar{a} \equiv (1, 1, 0)$ and $\bar{b} \equiv (0, 1, 1)$ is
(A) 1 (B) 3
(C) infinite (D) 2

29. If the plane $ax + by + cz = 1$ meets the coordinate axes in A, B and C then the centroid of the ΔABC is
(A) (3a, 3b, 3c) (B) $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$
(C) $\left(\frac{1}{3a}, \frac{1}{3b}, \frac{1}{3c}\right)$ (D) $\left(\frac{3}{a}, \frac{3}{b}, \frac{3}{c}\right)$

30. The degree and order of differential equation of family of circle touching a parabola $y^2 = 4\lambda x$ ($\lambda \in R$) at its vertex is
(A) degree = 2, order = 1 (B) degree = 1, order = 2
(C) degree = 1, order = 0 (D) none of these

Rough work